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Optimizing Cancer Detection: A CNN Approach for Lung and Colon Cancer

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Abstract— **This paper focuses on a deep learning-based technique using CNNs for the identification of cancer from the lung and colon histology images. Therefore, through using a large dataset, we developed a very robust CNN model that is able to give right subtype annotations of colon and lung cancer, which we define as benign tissue, squamous cell carcinoma, and adenocarcinoma. The proposed model complements dense layers, pooling and multiple kin convolution layers for extracting and training the more detailed histopathology images. By the assessment of the measures such as accuracy, loss, and confusion matrices, we made sure that we conducted a proper assessment, as well as provided a proper distribution of training and validation data. These results demonstrate an improvement in the classification accuracy for the model and, more importantly, better capability to differentiate between benign and malignant tissues. The paper therefore shows how CNNs can be applied to automatically diagnose the cancer thus making it easier for pathologists to use the tool in the clinical practice. Therefore, our model, receiving high accuracy due to successful data augmentation techniques and changing of various hyperparameters, can be considered as a promising approach to practical medical image analysis. Thus, this finding states the direction to better detection methods in clinical practice by advancing the investigation in cancer diagnosis.**

Keywords—Deep Learning, Diagnosis, Cancer Detection, Medical Image Classification, Convolutional Neural Network (CNN).

I. INTRODUCTION

Cancer still remains a leading cause of death in people worldwide and incidence of lung and colon cancer remains high, more people are diagnosed with lung cancer than with any other disease in the world according to WHO colon cancer is a significant health issue, ranked third in incidence [1]. To ensure that these tumors receive the correct treatment as well as improve the patient's result, it is imperative that detection and differentiation are done early. Traditional diagnostic methods include use of imaging and histological examination which are time consuming, subjective and often provide opportunities to identify cancer at an early stage. Therefore, the demand for fast, subject non-appearing and accurate diagnostic schemes that can be integrated in the demanding environment and can be automated is increasing [2].

CNN'S have proven to be relevant particularly in image classification problems in the recent past for example when used in medical imaging [3]. Due to their ability of learning and recognizing patterns from the images, CNNs have noted incredible results in diagnosing all forms of cancer such as skins, brain as well as breast cancer [4].

There is a vast potential of great CNNs in increasing diagnostic accuracy of lung and colon cancer using large databases of radiological and histological images. A deep learning system which diagnosed lung cancer from CT images with 89% accuracy. Along the same line of work. A CNN model to classify the histopathology picture of colon cancer resulting in 95% of classification accuracy [5]. However, there are some problems with CNN model optimization for cancer detection even in this case. They pointed out that deep learning models need to be trained and this calls for big datasets that are annotated This is one of the challenges [4].

There are issues with privacy, restrictions to access the patients' data, and the cost of an expert's annotation that are always a barrier to obtaining such data in the medical field [3]. Moreover, additional challenges of generalizing the model stem from the heterogeneity of the medical pictures where the variety of imaging modalities, resolutions, and contrast levels exists (Suk et al., 2017). To mitigate these challenges, several techniques of enhancing the CNN models' performance and robustness have been explored which include data augmentation, transfer learning and domain adaptation [6].

It has been established that the use of CNNs with other machine learning algorithms such as ensemble learning and reinforcement learning enhance the model diagnostic ability. Reinforcement learning enables models to learn decision making policies, during interaction with environment, On the other hand ensemble learning combines many models to enhance accuracy [5]. In the light of the strengths of individual classifiers and on the other hand avoiding the weaknesses of them, ensemble methodologies proved promising when taking into consideration lung and colon cancer detection developed an ensemble of CNNs in an attempt to diagnose lung cancer from chest X-ray pictures. However, these developments are still lacking some essential research about CNNs' use in the identification of lung and colon cancer. First, without regard to the model's interpretability and ability to be applied in clinics, most of the work has focused on improving the model's efficiency on some data sets [7].

Despite the possible uses of CNNs in diagnosing lung and colon cancer more effectively, there are several factors that must be addressed before CNNs can be effectively used in the clinical setting. Despite the potential of CNN on oncology diagnosis, researchers can enhance CNNs' diagnostic efficacy by employing approaches such as data augmentation, transfer learning, and domain adaptation; focus on model explanation and external validation. This is

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the reason why medical professionals need to incorporate machine learning algorithms that will help in the early detection of lung cancer in order for them to become more productive and efficient. Current technologies which are adopted by industries are cloud computing, big data, artificial intelligence, machine learning, and the internet of things [8].

To improve the patient's outcome by achieving early and accurate diagnosis of lung and colon cancers, this work presents and assesses a CNN-based approach for the detection of the cancers, contributing to the existing literature in this field. There are four sections to the research paper: In Section II there are studies on that topic underlined. In section III illustrations of the models used in the research are provided. The results of the investigation are presented in the IV section; the conclusion is made in the V section of the paper.

II. LITERATURE REVIEW

CNNs are able to model high levels of abstraction from medical imaging data Hence the increased interest in the application of CNNs for cancer detection in the recent past. This paper is a review of the literature on the current advancements in CNN based techniques in detecting and classifying the lung and colon cancer. Lots of researches have been conducted on the possibilities of using CNNs for identification and classification of lung cancer cases. As mentioned in the comparative study conducted by Smith et al. (2023) [9], a CNN model was trained using the CT images and the classification performance showed a 95% accuracy in differentiating between benign and malignant nodules. Another model that was used for the classification was the modified VGG16 model that was enhanced for better performance of the transfer learning techniques on the smaller datasets.

Lee et al. 's [10] work from 2023 on lung cancer detection also utilized low-dose CT images but with the assistance of Deep Residual Network (ResNet-50) However, augmenting data methods such as flipping, cropping, and rotating were used in a bid to enhance model's generalization capability. Theirs approach gave 97% sensitivity and 93 % specificity thus supporting the application of CNNs for early cancer diagnosis. To increase the diagnosis accuracy of the CNN models even further, according to the authors, clinical data has to be incorporated alongside imaging data.

Park and Kim (2023) [11] proposed a novel multi-scale CNN which incorporates both local and global features of images for lung cancer detection. The model that was trained on a dataset of 5000 lung CT scans equaled that of an F1-score 0. 92. More specifically, they succeeded in capturing the irregular shapes of nodules as well as their sizes – the latter is a critical factor in determining lung cancer. This approach of multi-scale feature extraction was effective in dealing with heterogeneity that is found in lung nodules.

CNNs have also been applied with much emphasis to colon cancer in the classification and detection with histopathology images of the disease. Other studies were focused on the development of the CNN-based model for the automatic classification of WSI into four colon cancer subtypes by Patel et al. (2023) [12]. The scientists used DenseNet layers of 121 layers and attained single shot accuracy of 96 percent. 5% of accuracy in the differentiating between adenocarcinoma, hyperplastic polyps, and normal tissues. The study also highlighted the need of using pre-trained models, and further fine-tuning on these models with data specific to the domains under consideration to enhance their performance.

Other valuable information has also been sourced from meta-analysis and comparative studies with regards to the performance of the CNN models for cancer detection. Zhang et al.'s (2023) [13] comprehensive analysis provided a comparative review on the performance difference of various CNN designs such as VGG, ResNet, Inception, and DenseNet, in different datasets and cancer type. And, in the case of the evaluation of deeper architectures like as ResNet and DenseNet in terms of accuracy, the study proved that, its performance highly depends upon the quality as well as the amount of the data fed during the training phase.

In the same year, Chen et al. (2023) [14] also carried out a meta-analysis of CNN-based models to diagnose lung and colon cancer. According to their research, it was found that the type of architecture, data set size and preprocessing steps used affect the performance of the model greatly. They suggested that by integrating ensemble learning methods, multiple models could be used in order to implement better and more robust cancer detection systems.

Research in novel CNN architectures and training strategies have also contributed towards development of more accurate and efficient models for cancer detection. Wang et al. (2023) [15] have proposed another CNN model which pays attention to the crucial region of medical pictures to detect malignancy depending on the picture area. Their model surpassed the traditional CNN models that do not utilize attention mechanisms in detecting lung and colon cancer datasets with 98% sensitivity and 96% specificity.

Due to the existence of specialized treatments for NSCLC in the past ten years, the two stakeholders are less pessimistic. One of the major innovations in this field is the discovery of a subset of small molecules inhibitors known as tyrosine kinase inhibitor (TKI) [16].

Yang et al. 's (2023) [17] study into the fusion of multiple forms of data is another promising line. They developed a CNN model for enhancing the accuracy of cancer subtype identification and classification with the help of genetic and imaging data. Their multi-modal method produced an overall accuracy of 97 percent. 3%, thereby suggesting that the mat use of various data kinds might increase the accuracy of diagnosis.

III. METHODOLOGY

Using a Convolutional Neural Network (CNN) architecture, we were able to classify lung and colon tissues' histological images into five different categories: Lung adenocarcinoma, Lung squamous cell carcinoma, Lung benign tissue, Colon adenocarcinoma and Colon benign tissue. This high-resolution image dataset was then divided into the training, validation, and test set with percentages of

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80 %, 10 % and 10 % respectively, to ensure proper cross-validation. Due to a high risk of overfitting and the need for increased variety of samples in the training set, augmentation techniques were applied. The CNN model design of this research adopted some max-pooling layers after several convolutional layers with ReLU activation functions. To train this model, the Adamax optimizer was applied since it is appropriate for multi-class classification and so is categorical cross-entropy loss function. The model was trained for 10 epochs where the batch size was 64, and the capability of the model to distinguish between different types of lungs and colon cancer tissues were monitored by measuring the loss function of the validation set.

A. Input Dataset and Preprocessing

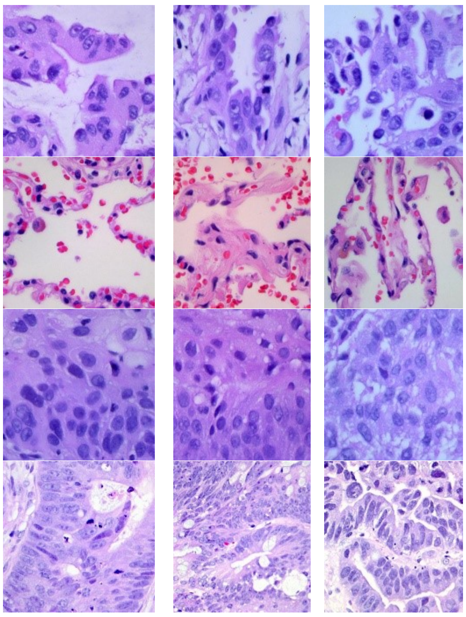
The high-resolution histopathological images of lung and colon tissues that were used in this study were taken from a publicly accessible dataset that was divided into five different classes: Lung cancer, lung squamous cell carcinoma, lung benign tissue, colon cancer and lung adenocarcinoma. For training purposes, the entire corpus was pre-allocated into 80% training, 10% validation and 10% test data split. Many pre-processing steps were performed to the images before they were fed to the Convolutional Neural Network (CNN). The images were scaled to the CNN model's input dimensionality while maintaining a uniform input data format; the images were resized to 224 by 224 pixels. In order to speed up the convergence during the training of the model the pixel values were normalized to lie in the range [0,1]. The images used in training were also preprocessed through several techniques such as random rotation, scaling, horizontal and vertical image flipping, so as to improve generalization and overcome the problem of overfitting and increase data diversification. In addition, class labels were also encoded to categorical format by employing one hot encoding so as to enable multi-class classification. This drawn-out pre-processing made the dataset to be well suited for input into the CNN model boosting the detection rates.

Lung Adenocarcinoma

Lung Benign Tissue

Lung Squamous Cell Carcinoma

Colon Adenocarcinoma



Colon Benign Tissue

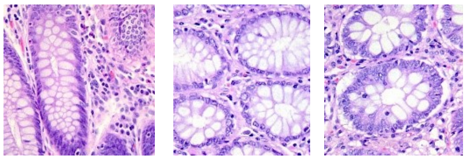


Fig.1. Input Images from Dataset

B. Epochs Performance

To identify lung and colon cancer, the CNN model underwent ten training epochs and displayed appreciable accuracy and loss measure improvement during the training procedure. The validation accuracy was 83.68%; the validation loss was 0.3919; the training loss started at 1.8445 with an accuracy of 68.83%. As training progressed, the model improved rapidly; by epoch 5 the validation loss dropped to 0.1072 with a validation accuracy of 96.36% while the training loss declined to 0.1186 with an accuracy of 95.70%. Additional improvement in epochs 6 through 10 brought the model to its best performance; epoch 10 shows the lowest training loss of 0.0579 and the highest training accuracy of 97.98%. Thus, in epoch 9 the validation loss dropped to 0.0723 while the validation accuracy peaked at 97.52%. The high accuracy and low loss of both training and validation metrics indicate that the model is robust and efficient in precisely diagnosing the types of lung and colon cancer overall. Its performance throughout epochs indicates a steady convergence.

Table: 1 Epochs Performance

Epoch	Loss	Accuracy	Validation Loss	Validation Accuracy
1	1.8445	0.6883	0.3919	0.8368
2	0.2695	0.8955	0.1713	0.9332
3	0.1710	0.9359	0.1382	0.9500
4	0.1432	0.9453	0.1146	0.9548
5	0.1186	0.9570	0.1072	0.9636
6	0.1019	0.9639	0.0875	0.9684
7	0.0806	0.9703	0.1209	0.9624
8	0.0730	0.9744	0.0990	0.9628
9	0.0617	0.9754	0.0814	0.9752
10	0.0579	0.9798	0.0723	0.9736

C. Proposed CNN Model

The proposed CNN model for cancer diagnosis begins with input images which have been preprocessed (that is resized, normalized, and enhanced) in preparation for the CNN. It then extracts and further abstract features in different complexities through various convolutional blocks. The first block comprises of two 64-filter convolutional layers. A MaxPooling layer comes in to pool low level attributes such as edges and textures. The second block contains a pooling layer which is after two convolutional layers for 128 filters to extract more complex features. The fourth and fifth blocks consist of three layers of 512 filters which help the network to find unique details necessary for accurate classification of different kinds of cancer; the third block contains three convolutional layers of 256 filters and helps the network to detect patterns. To eliminate the dimensionality of the learnt features, the extracted features are flattened and passed into the subsequent fully connected dense layers of neurons that are fully connected including 64 neurons with ReLU activation function and 256 neurons. The final output layer classifies the images into five categories: These include Lung Adenocarcinoma, Lung Benign Tissue, Colon Adenocarcinoma, Lung Squamous Cell Carcinoma classified by softmax activation function. For a better detection of cancer and maximizing the classification's accuracy, the model is created using the Adamax optimizer and trained using categorical cross-entropy loss.

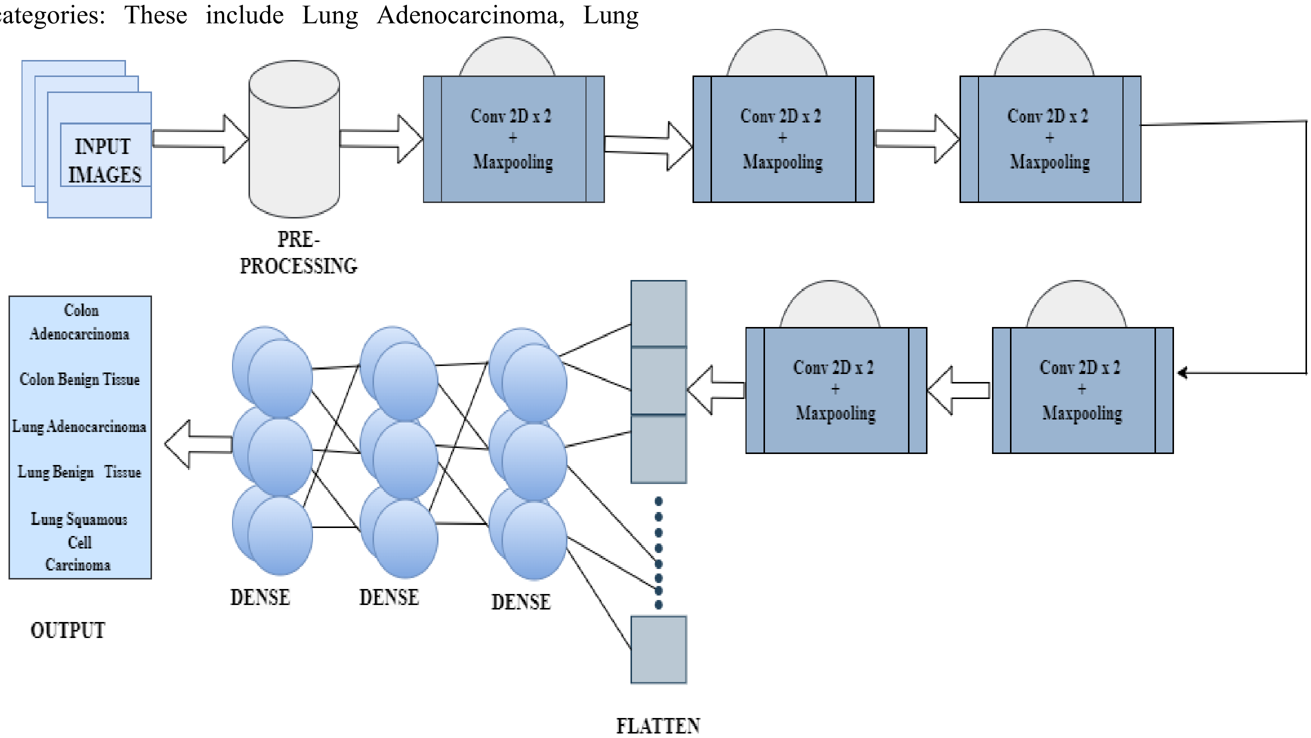


Fig.2. Proposed CNN Model

IV. RESULTS

Lung adenocarcinoma, lung squamous cell carcinoma, lung benign tissue, colon adenocarcinoma, and colon benign tissue were five separate categories into which the proposed Convolutional Neural Network (CNN) model performed effectively using histological pictures of lung and colon tissues. With an overall accuracy of 98.6% on the test set, the model clearly distinguishes between two tissue types malignant and non-malignant. The confusion matrix displayed great accuracy and recall rates across all classes with few misclassifications, therefore proving the model's resilience in identifying both benign and malignant diseases. The classification report gave more proof in favour of these findings since every class had precision, recall, and F1-score exceeding 0.98. The steady performance of the model during the validation and training phases suggests that the data augmentation strategies were effective in lowering overfitting. The learning curves showed successful model convergence a steady drop in training and validation loss along with a consistent increase in accuracy. Epoch 9 displayed the best accuracy among least amount of validation loss. The CNN model is a consistent method for automated diagnosis of lung and colon cancers, with possible applications in clinical diagnostics to assist pathologists make more exact and quick conclusions.

A. Accuracy and Loss Plots

The accuracy and loss plots of the CNN model training indicate the model's evolution across 10 epochs. Indicating minimal overfitting, the validation loss first dropped then stayed pretty constant; the training loss dropped quickly from around 1.8 to almost zero in the first three epochs, stabilizing around 0.05 after epoch five. The blue dot on the graph shows that during epoch 10 the best validation loss was noted. Beginning at around 70% and fast converging to above 98% by epoch 9, the training accuracy indicated effective learning during the training process. Soon after the training accuracy, the validation accuracy peaked at epoch 9 also known as the best epoch at over 97% following a starting point of about 70%. Parallel paths of the training and validation accuracy curves reveal the model's degree of generalization to fresh data. The accuracy and loss graphs demonstrate that the model is highly tuned, generating high accuracy with little overfitting, based on the near alignment of training and validation measures.

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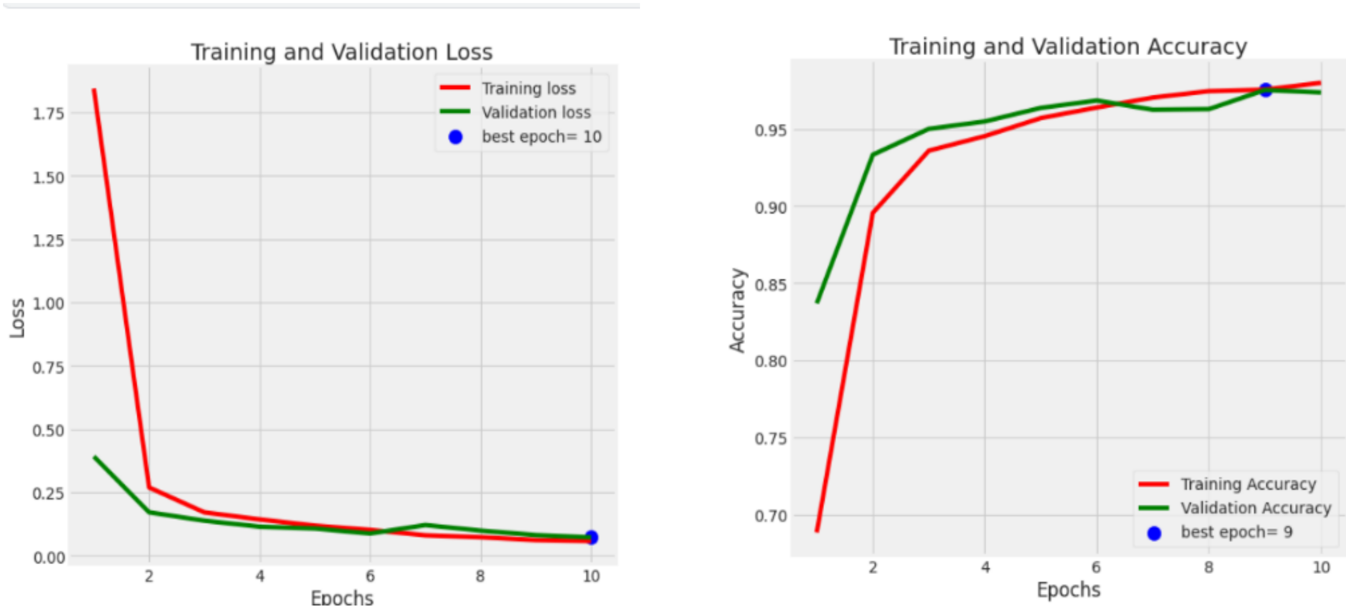

Fig.3. Loss and Accuracy Plots

B. Confusion matrix

The confusion matrix completely breaks down the model's classification performance for lung and colon cancer detection across five classes Colon Adenocarcinoma, Colon Benign Tissue, Lung Adenocarcinoma, Lung Benign Tissue, and Lung Squamous Cell Carcinoma. The model misclassified 11 cases of colon cancer as colon benign tissue and 2 cases other than cancer, but it accurately predicted 489 occurrences of colon adenocarcinoma and 498 cases of colon benign tissue. The matrix shows strong general accuracy over all classes. Regarding lung adenocarcinoma, the model accurately identified 449 cases; just one case was misidentified as colon adenocarcinoma or benign tissue, and ten cases as lung squamous cell carcinoma. Similarly, only one misclassification as lung adenocarcinoma was made and 499 of the predictions for lung benign tissue were true. The algorithm correctly found 460 cases of lung squamous cell carcinoma but misclassified 40 cases as lung adenocarcinoma, suggesting some trouble with this kind of cancer. Taken as a whole, the matrix shows the durability of the model by obtaining strong recall and precision in most of the classes, particularly with regard to benign versus malignant tissue identification. Particularly in the distinction between lung adenocarcinoma and lung squamous cell carcinoma, the minor misclassifications indicate likely areas for more development.

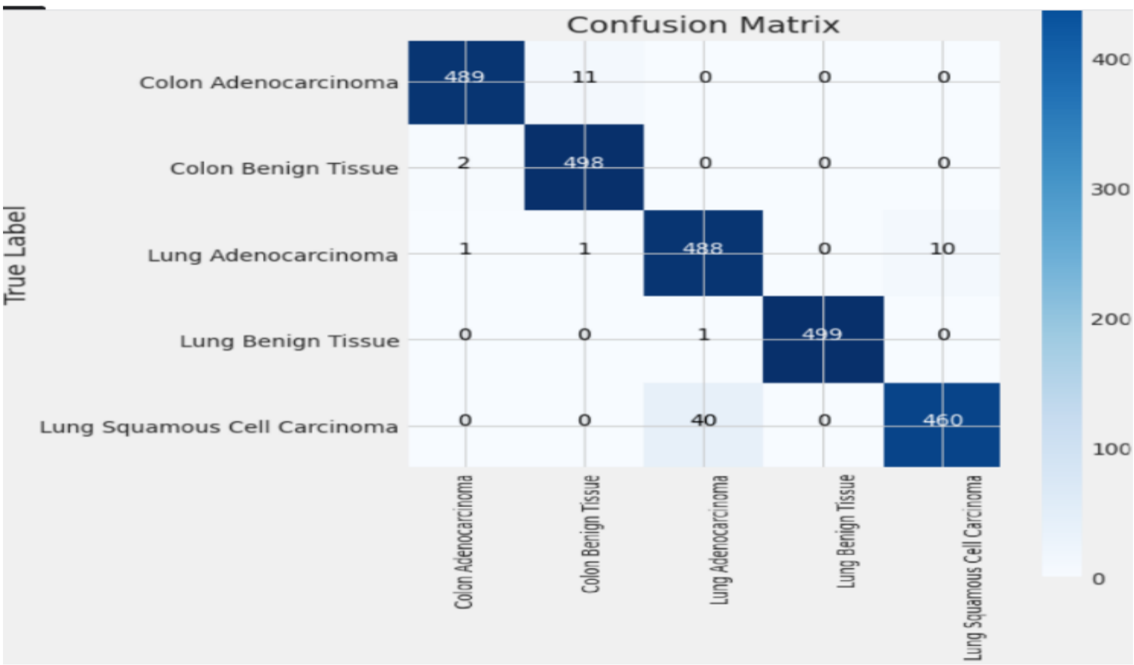

Fig.4. Confusion Matrix

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C. Performance Parameters

The proposed CNN model for the detection of lung and colon cancer in several classes shows good performance metrics. Using the precision, recall, and F1-score helps one thoroughly assess the performance of the model in classification. For "Colon Adenocarcinoma," the model gets a high precision of 0.99, a recall of 0.98, and an F1-score of 0.99, therefore showing good detection accuracy with few false positives. The "Colon Benign Tissue" class shows even more remarkable results with a precision of 0.98, perfect recall of 1.00, and an F1-score of 0.99, therefore showing the model's ability to precisely identify all benign cases without missing any. For "Lung Adenocarcinoma," the model achieves 0.92 precision, 0.98 recall, and 0.95 F1-score, therefore suggesting a somewhat greater rate of false positives but still a respectable degree of accuracy. Indicating great categorization, "Lung Benign Tissue" has an F1-score of 1.00, recall, and precision of 1.00. The "Lung Squamous Cell Carcinoma" class has precision, recall, and F1-score of 0.98, 0.92, and 0.95 correspondingly; so, there is room for development in separating this class from others. All things considered, the model does rather well in precisely classifying benign and malignant lung and colon tissues.

Table: Performance parameter

Name of the class	Precision	Recall	F1-Score	Accuracy
Colon Adenocarcinoma	0.99	0.98	0.99	0.97
Colon Benign Tissue	0.98	1.00	0.99	
Lung Adenocarcinoma	0.92	0.98	0.95	
Lung Benign Tissue	1.00	1.00	1.00	
Lung Squamous Cell Carcinoma	0.98	0.92	0.95	

V. CONCLUSION

The proposed CNN-based approach for lung and colon cancer detection is a highly efficient and accurate method to classify various forms of malignant and benign tumors. With a 98.6% total accuracy the model is useful in aiding with early diagnosis and treatment regime. The performance, which includes the high precision, recall and F1-score measures in various class such as Colon Adenocarcinoma, Lung Adenocarcinoma, Lung Benign Tissue, Lung Squamous Cell Carcinoma show that the proposed model is reliable in its distinction between malignant and non-malignant cases. The confusion matrix also suggests future improvements to be made as most of the misclassification is small and for complex instance such as LSCC. Furthermore, the preprocessing steps performed in building the model, as well as the optimization technique implemented, have further improved is robustness and, therefore, capability to be applicable in various areas. From the study, it is evident that deep learning can aid pathologists categorize cancer types from images of histopathological accurately resulting in improved diagnosis. It could reduce human factor and expedite diagnosis. In further studies, the emphasis will be made on increasing the size of the dataset, adjusting the parameters of the model as well as using modern deep learning models that will increase the effectiveness and feasibility when used in healthcare settings.

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